Robotics and Flexible Automation

Lecture 1
Subject introduction and robot selection

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BJTU · 2026

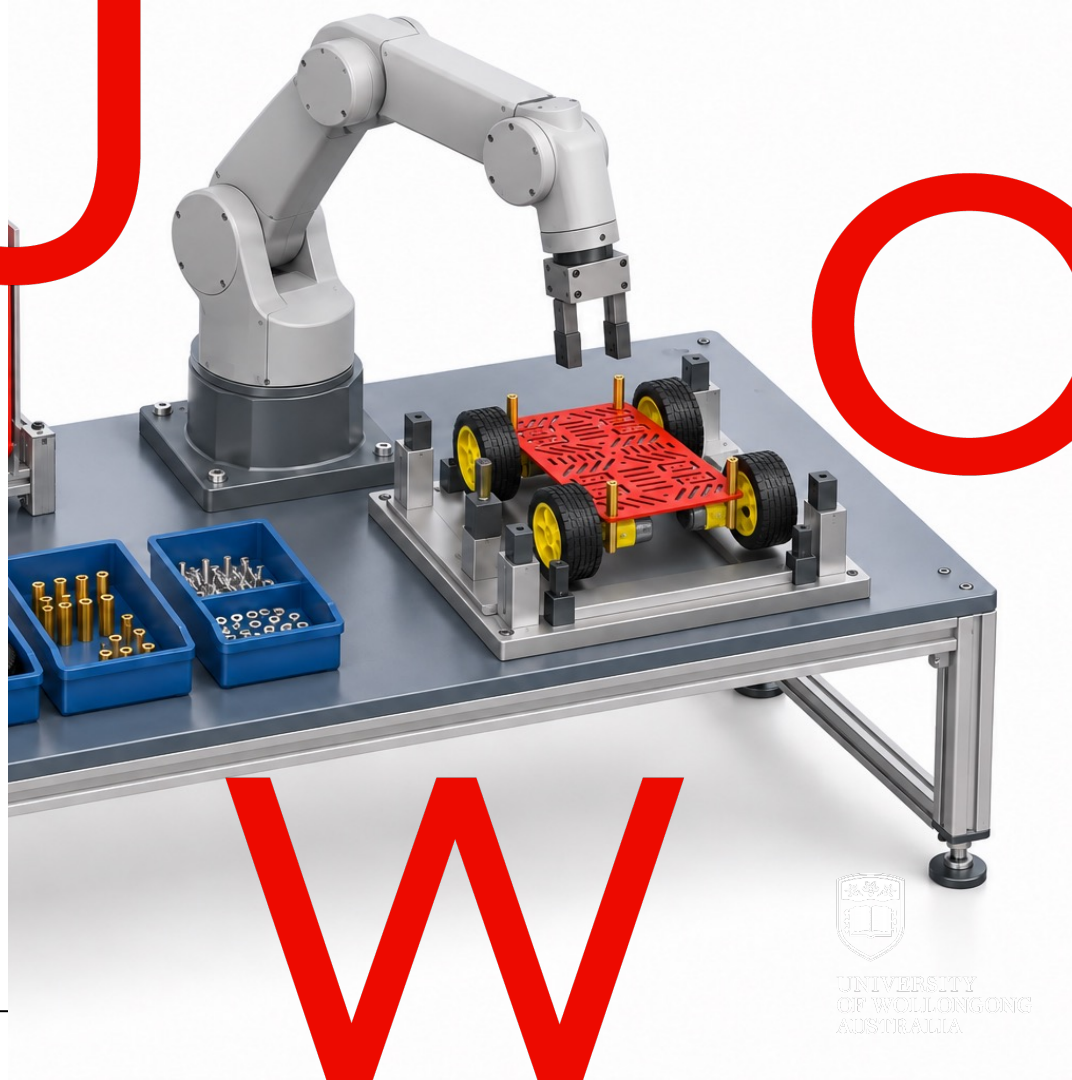


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PART 1

Subject orientation and learning tools

The course, the software and the support available

1. Learning outcomes and subject roadmap
2. Textbook and software choices
3. Resources and learning support

What you will learn from this subject

CORE

Understand the system → **Describe the tool pose** → **Build a kinematic model** → **Validate, plan and program**

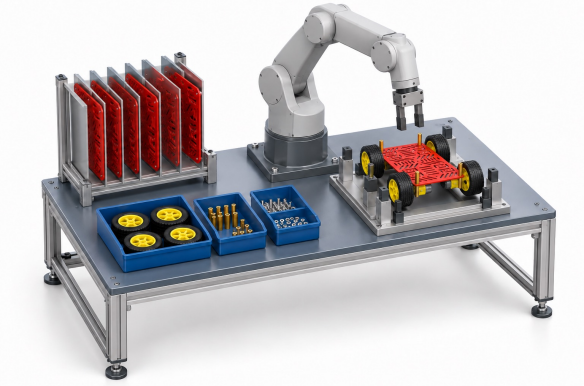
manipulator, controller, tool, fixture, sensors

where the tool is, and which way it points

joint angles ↔ tool pose

simulate, then run it on a real controller

mathematics → software model → a real machine doing the task

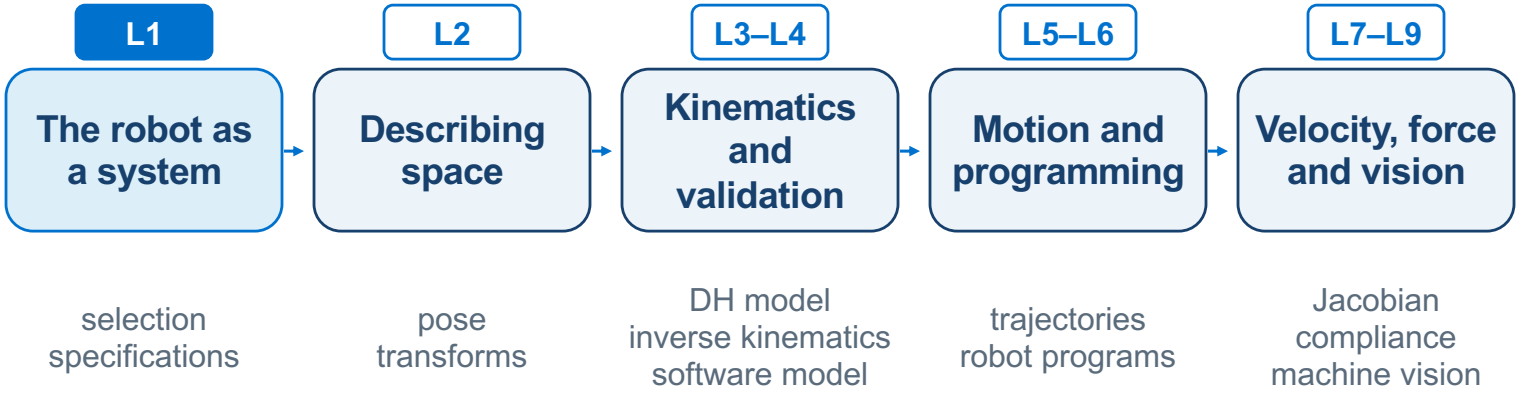


The assembly project

Plan ten placements: the lower panel, four wheels, four spacers and upper panel. Use modelling and simulation to justify the robot, end-effector strategy and motion.

The subject roadmap

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Each block assumes the one before it

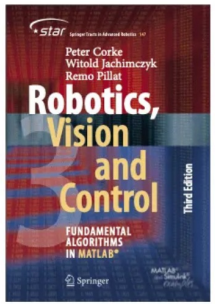
Project stages: concept and EOI, kinematic model and PDR, software validation and FDR, then integrated assembly and final tender.

The textbook

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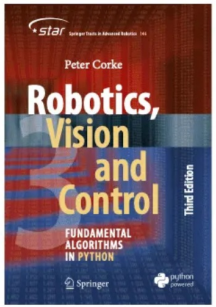
https://petercorke.com/books/robotics-vision-control-all-versions/

3e



All MATLAB (closed)

MathWorks



All Python (open)

python

2023

Peter Corke: Robotics, Vision and Control

MATLAB course examples: second edition + RTB10.

Python examples: third edition + Corke Python toolbox.

The MATLAB third edition uses MathWorks robotics software, NOT our coding route.

Use its theory only

The software tool: MATLAB or Python

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PYTHON — free, open source

pip install rvc3python

Corke robotics, spatial math and vision

- roboticstoolbox-python
- spatialmath-python
- machinevision-toolbox-python
- bdsim

MATLAB + Corke RTB10

Install RTB.mltbx in MATLAB

GitHub: RTB + Spatial Math + Common

- spatialmath-matlab
- toolbox-common-matlab

✕ Robotics System Toolbox

NOT USED IN THIS COURSE

Both course routes use Peter Corke toolboxes. MATLAB requires a licence; Python is free. Their APIs are different.

Corke MATLAB and Python

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A pose, a robot model, and its forward kinematics

PYTHON · Corke roboticstoolbox

```
from spatialmath import SE3
from math import pi
import roboticstoolbox as rtb

T = SE3(0.5,0,0) * SE3.Rx(pi/2)
T.plot()

robot = rtb.models.DH.Puma560()
robot.fkine(robot.qz)
```

MATLAB · Corke RTB10

```
T = transl(0.5,0,0) * trotx(pi/2);
trplot(T);

mdl_puma560        % creates p560
Tz = p560.fkine(p560.qz);
Tz.T               % inspect the 4 x 4
% metres; radians unless stated
```

Same concepts, separate APIs: build a pose, load a Corke robot model, and check forward kinematics.

Free resources, and how to get unstuck

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Everything on this slide is free and open to you

Robot Academy

robotacademy.net.au

Corke's free video library: over 200 lessons of under ten minutes, grouped into hour-long masterclasses. Built from the QUT robotics MOOCs.

Corke code and installation

github.com/petercorke

MATLAB: RTB10.
Python: Corke toolbox.
RVC3-python: book examples.
RVC3-MATLAB: not used.

The documentation

petercorke.com/toolboxes

Function reference for each toolbox — where you check whether an angle is in degrees or radians.

When something does not work: read the error, check the units and the convention, run the toolbox's own example, then ask.

Responsible use of AI

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AI may support your work. You remain responsible for the result.

Permitted support	Brainstorm ideas, clarify concepts and get feedback. Use AI to help with MATLAB/Python coding and debugging.
Understand and verify	Review and test every AI-assisted result. Check equations, robot specifications and references against reliable sources.
Your own authorship	Write the final report in your team's own words. Do not submit substantial AI-generated or AI-rewritten passages. Explain your work.
Declare and retain evidence	Include an AI Use Declaration: tool, purpose, where it helped and how you verified it. Keep drafts and development records.

Three Elements of Stupidity

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1 Outsourcing critical thinking to a third party
“Don’t use AI to do a draft.”

2 Willful ignorance. (Amathia)
“I do not know, and I do not want to find out.”

3 Conformity
“Everyone in my group does it, so I will too.”

Jonny Thomson · Philosopher, author and Big Think columnist
Adapted from: The 3 elements of stupidity, according to philosophy (Big Think, 2026)

PART 2

Robot structure

The physical robot, its joints and its task variables

1 Robot system and anatomy
2 Degrees of freedom, TCP and pose
3 Common architectures

The industrial robot system

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An open teaching cell: robot, tooling, workpiece and control hardware



Source: ABB Robotics Education Package (2023), GoFa teaching cell. Labels added.

Robot anatomy and joint types

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Links are rigid members. Joints permit relative rotation or translation.

Links and joints

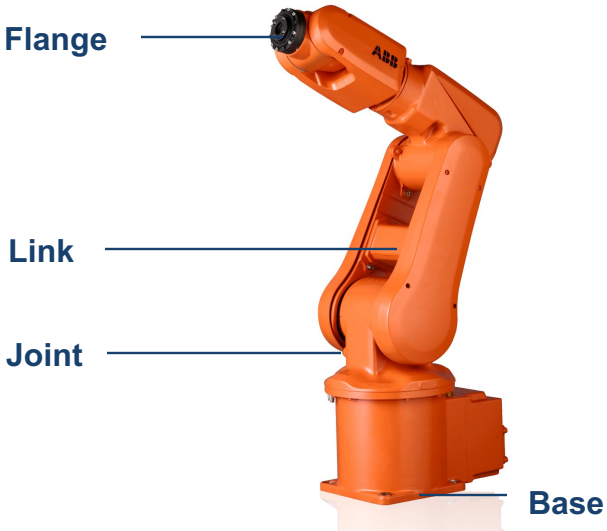


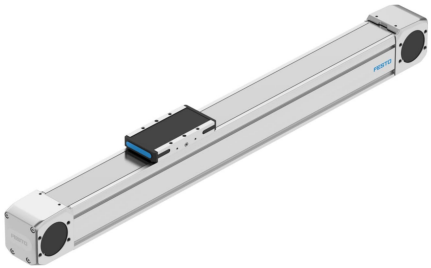
ABB IRB 120

Revolute joint (R)



Rotation about an axis
Joint variable: angle θ

Prismatic joint (P)



Translation along an axis
Joint variable: distance d

Sources: ABB / Cision (IRB 120). Festo / DirectIndustry (ERMO). Festo / RS (ELGD).

Degrees of freedom

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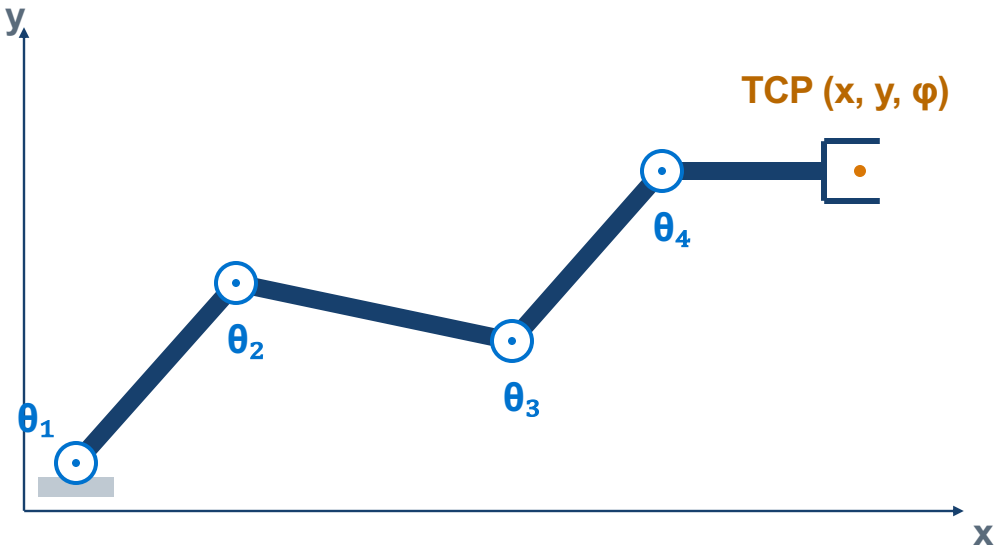
A planar 4R arm: four joint variables, three tool-pose coordinates

Four revolute joints

Each joint rotates about an axis perpendicular to the page.

Tool pose in the plane

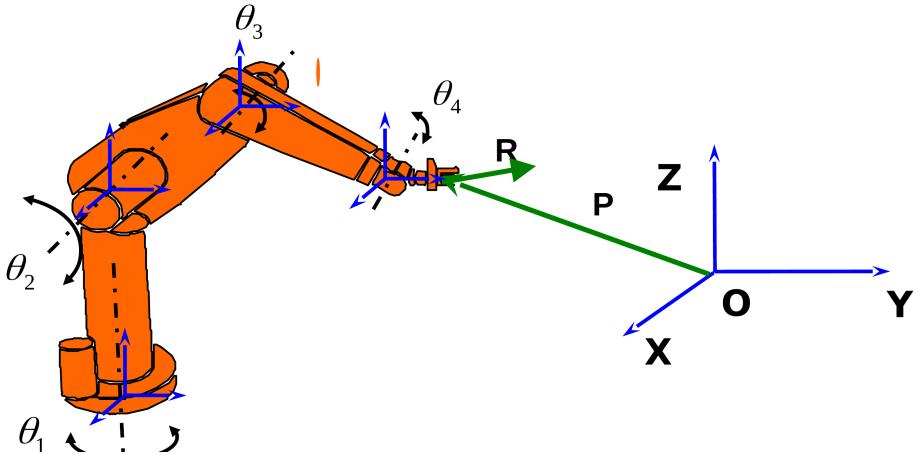
Position: x, y
Orientation: φ
No out-of-plane translation or tilt.

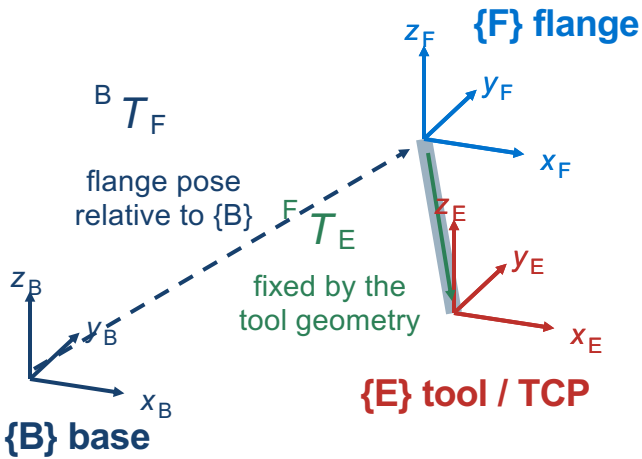


4 joint DOF → up to 3 planar tool-motion DOF. The extra joint permits alternative configurations.

Joint space, tool pose and the TCP

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Joint space

q = (q_1 … q_n) — what the motors and encoders control.

Tool pose

Where the tool is and which way it points — how the task is written.

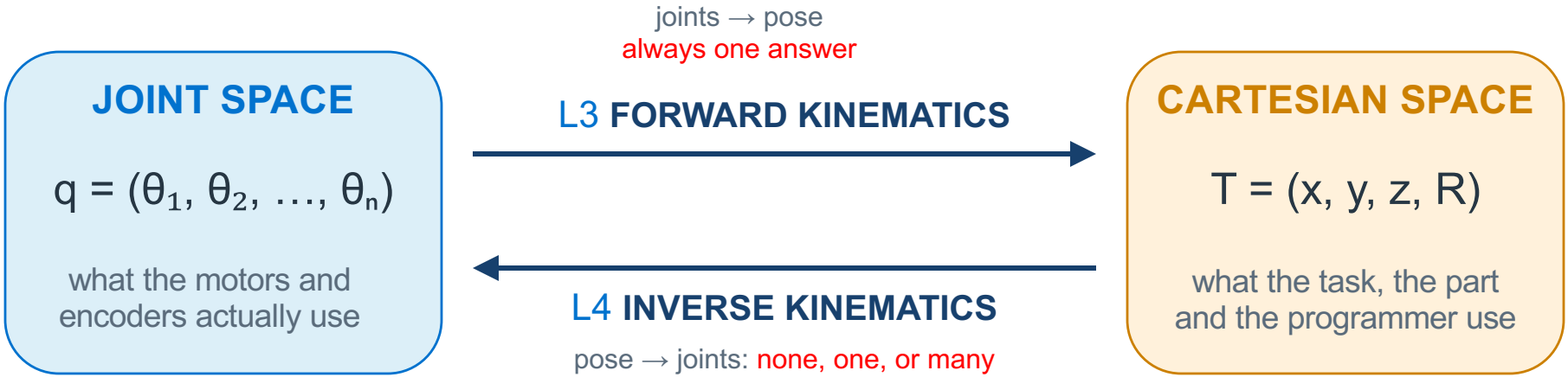
Flange, tool, TCP

The tool frame is fixed relative to the flange. Its origin is the TCP.

Forward and inverse kinematics

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Joint coordinates and tool pose are linked by the robot model (DH model)



Common robot architectures

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Real robot architectures and typical examples

Articulated



ABB IRB 120

Serial rotary joints
Flexible tool orientation

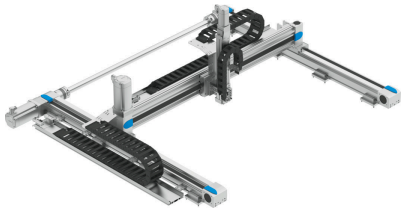
SCARA



Epson SCARA

Planar motion + vertical axis
Vertical insertion

Cartesian



Festo gantry

Orthogonal linear axes
Large rectangular workspace

Delta



ABB IRB 360

Parallel linkage
Rapid handling of light parts

Images: ABB / ManualsLib; Epson; Festo; ABB / A3 Automate. Product examples shown at different scales.

Industrial robot manufacturers

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Manufacturer	Origin	Industrial robot examples	Typical applications
ABB	Sweden	IRB series	Handling, welding, assembly
KUKA	Germany	KR series	Automotive, handling, assembly
FANUC	Japan/USA	M / R / LR Mate	Handling, welding, assembly
Yaskawa/Motoman	Japan	GP / MH / AR	Handling, welding
Kawasaki	Japan	RS / BX	Handling, spot welding
Epson	Japan	G / LS SCARA	Small-parts assembly
Stäubli	Switzerland	TX / TS	Precision assembly, handling
Universal Robots	Denmark	UR series cobots	Flexible assembly, machine tending
Techman Robot	Taiwan	TM series cobots	Vision-guided assembly, handling
Siasun 新松	China	Industrial robot arms	Welding, handling, assembly
ESTUN 埃斯顿	China	ER / UNO	Welding, handling, assembly
EFORT 埃夫特	China	Industrial articulated robots	Painting, welding, handling
Inovance 汇川	China	SCARA / six-axis	Electronics assembly, handling

Sources: manufacturers' official product ranges. Cobot suppliers highlighted in green; Chinese suppliers in blue.

Humanoid robots

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A different architecture for tasks in human-designed environments

1. **Whole-body motion**

   Balance, walking and arm motion must be coordinated continuously.

2. **Programming changes**

   Task-level commands, teleoperation and demonstration learning complement point-to-point teaching.

3. **Perception and contact**

   Vision, force and tactile feedback are needed in less structured environments.

4. **New engineering challenges**

   Fall recovery, battery life, actuator heat, dexterous hands and safe contact.



Unitree G1

Best fit: variable tasks in spaces designed for people. For a fixed workcell, a simpler robot is usually faster, cheaper and easier to validate.

PART 3

Robot specification and selection

Turning an application into a defensible robot choice

1. Datasheets, reach and workspace
2. Payload, joint limits and repeatability
3. Calibration, end effectors and selection priorities

Reading a robot datasheet

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ABB IRB 2600 — enlarged original specification extracts

Specification

	Variants	Reach	Payload	Armload
1	IRB 2600-12/1.65	1.65	12	15
	IRB 2600-20/1.65	1.65	20	10
	IRB2600-12/1.85	1.85	12	10
2	Number of axes	6+3 external (up to 36 with MultiMove)		
3	Protection	Standard IP67; optional FoundryPlus 2		
	Mounting	Floor, wall, shelf, tilted, inverted		
	IRC5 Controller variants	Single cabinet, Dual cabinet		
	Physical			
	Dimensions robot base:			676 x 511
	Robot height: IRB 2600-12/1.65 and IRB 2600-20/1.65			1382 mm
	Robot height: IRB 2600-12/1.85			1582 mm
4	Robot weight:			272 to 284 kg

Performance (according to ISO 9283)

		Arm 1.65	Arm 1.85
5	Position repeatability (RP)	0.04 mm	0.04 mm
	Path repeatability (RT)	0.13 mm	0.16 mm

Movement

	Axis movements	Working range	Maximum speed
6	Axis 1	+ 180° to -180°	175 °/s
	Axis 2	+ 155° to - 95°	175 °/s
	Axis 3	+ 75° to - 180°	175 °/s
	Axis 4	+ 400° to - 400°	360 °/s
	Asix 5	+ 120° to -120°	360 °/s
	Axis 6	+ 400° to - 400°	500 °/s

Read the model variant first.

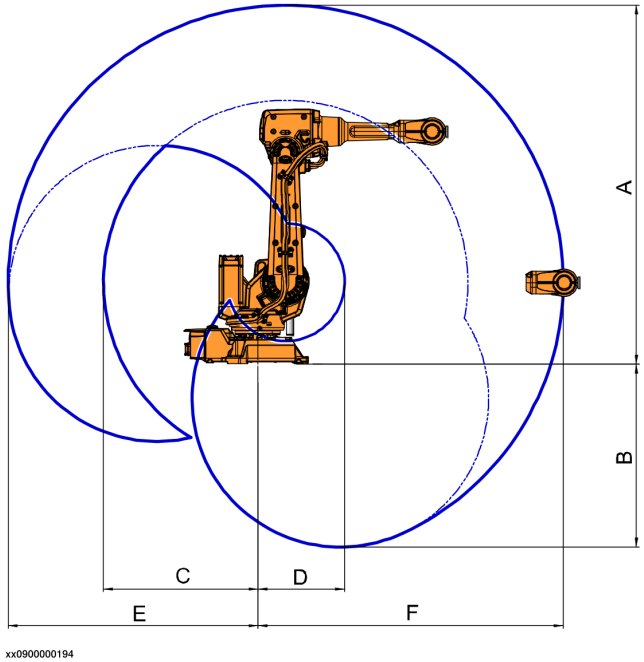
1 Reach, payload, armload
2 Axes and external axes
3 Protection / mounting
4 Robot mass
5 Repeatability
6 Joint range / speed

Source: ABB IRB 2600 datasheet supplied with this lecture. Original table crops; numbered annotations retained.

7 Working range: see next slide.

Reach and workspace

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Position reachable ≠ required pose reachable

A point may be reachable only with certain joint configurations and tool orientations.

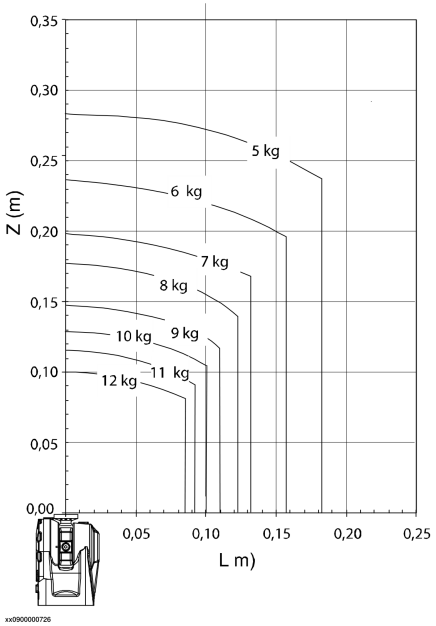
The usable region for a real job can be much smaller once tool orientation, approach path and collision clearance are required.

Source: ABB IRB 2600 Product Manual, 3HAC035504-001 Rev. AK, p. 51. Envelope measured at wrist centre.

Payload and tool loading

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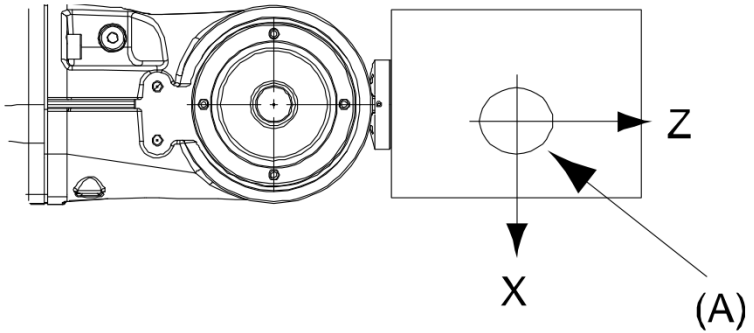
Payload depends on both mass and where that mass is carried



Extra load of 15 kg at the upper arm housing and 1 kg at wrist included in the load diagram.

Load diagram "Vertical wrist" is not available for IRB 2600-12/1.65.

Include tool + adapter + workpiece



A = centre of gravity; L = offset from the wrist axis.

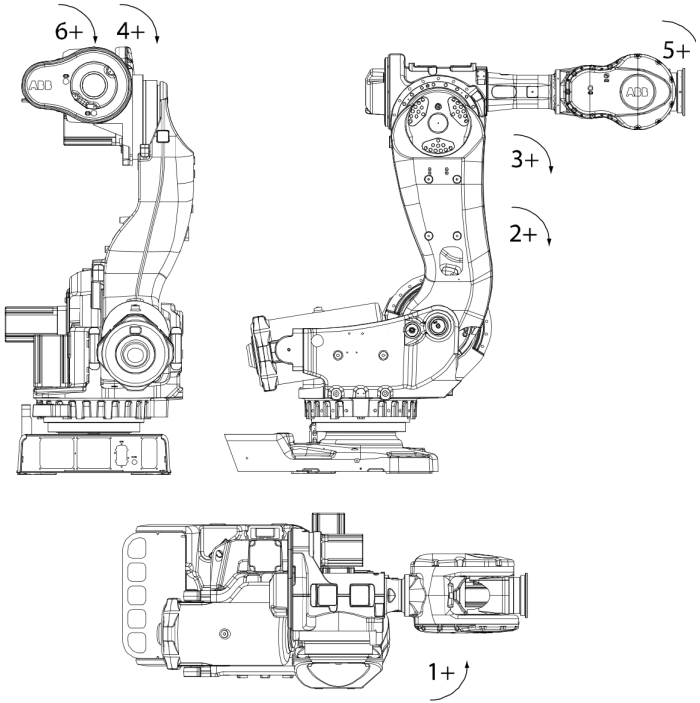
Check the load curve and inertia limits. A nominal 12 kg payload is not valid at every centre-of-gravity offset.

Source: ABB Product Specification, 3HAC035959-001 Rev. AG, pp. 43 and 50. Example: IRB 2600-12/1.65.

Joint limits and configuration

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Joint limits constrain which configurations the robot can use



IRB 2600 axis ranges

Axis	Type of motion	Range of movement
1	Rotation motion	+ 180° to - 180° [i]
2	Arm motion	+ 155° to - 95°
3	Arm motion	+ 75° to - 180°
4	Rotation motion	+ 400° to - 400° + 251 rev. to - 251 rev. Max.
5	Bend motion	+ 120° to - 120°
6	Turn motion	+ 400° to - 400° + 274 rev. to - 274 rev. Max.

A reachable position can still fail because a required configuration exceeds a joint limit.

Sources: ABB Product Manual Rev. AK, p. 415 (generic IRB 7600 axis illustration); Product Specification Rev. AG, p. 64.

Resolution, repeatability and accuracy

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RESOLUTION

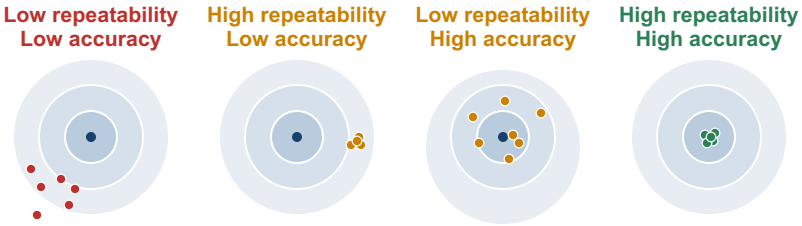
— the smallest step the controller can command

REPEATABILITY

— how tightly it returns to the SAME taught point — 0.04 mm for the IRB 2600

ACCURACY

— how close the attained pose is to the commanded pose



Repeatability = how tightly the shots cluster.
Accuracy = how close the cluster centre is to the target.



the smallest commanded step the controller can produce — not the same thing as accuracy

A taught program relies on repeatability. A program generated from CAD coordinates relies on accuracy — which is usually much worse.

Robot calibrations

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Three levels connect joint sensing, the tool and external measurements.

1 Robot joint reference	**ABB revolution counter update** Synchronize encoder/resolver turn counts with known joint positions. This restores the joint reference. Fine calibration is a separate procedure.
2 Tool / TCP calibration	**ABB tool-definition routine** Bring the tool tip to one fixed point from several orientations. Estimate the TCP relative to the flange, then check the residual error.
3 External sensors	**Camera and other measurement devices** Calibrate camera intrinsics and its pose relative to the robot. Other sensors may need a zero, scale or mounting-transform calibration.

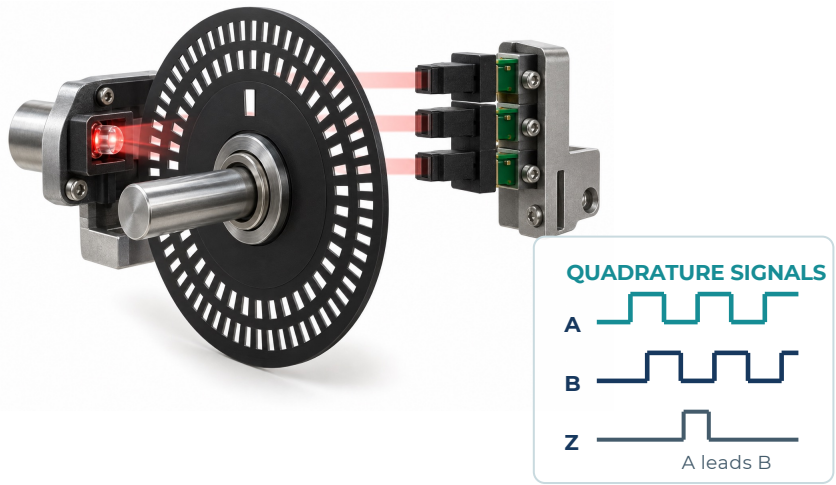
Later: ABB tool setup in Lecture 6; camera calibration in Lecture 8.

Incremental encoder and joint reference

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Why robot joints need a known reference position

A outer incremental track
B offset incremental track
Z one index per turn



What the controller knows

A/B phase gives direction.
Pulse rate gives speed.
Count gives relative position.
Z marks one angle per motor turn.

$$\Delta\theta = \frac{2\pi}{N}$$

N = decoded counts per revolution.
The same Z pulse repeats every turn; a retained turn count is still needed.

Joint reference: restore the turn count at a known joint position. ABB calls this a revolution counter update.

End-effector selection

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Select and justify an end-effector strategy for all ten placements

Parallel gripper



Grip rigid parts with two fingers

Vacuum gripper



Pick flat, non-porous surfaces

Magnetic gripper



Hold ferrous sheet or parts

Adaptive gripper



Conform to irregular shapes

Images: original lecture sources; OnRobot MG10; Robotiq 3-Finger Adaptive Robot Gripper. Application matches are illustrative.

Robot selection priorities

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Ask	Evidence you must show
Does it fit the workcell?	Base footprint, height, mounting and service clearance
Can it reach every operation?	Pick, approach and assembly poses, with base and TCP defined
Does it have enough freedom of motion?	Axis count and wrist orientation for the required approach directions
Can it manoeuvre without collisions?	Access around the fixture, product and cell structure
Is the tool and part load acceptable?	Check payload and centre of gravity after the geometry works
Is it a sensible choice?	Why this architecture and size fit the task

Start with geometry: size, reach and degrees of freedom decide whether the operation is feasible. Check payload and joint limits after shortlisting models.


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